Control System Training

MODULE 10 – Analog Control Concepts

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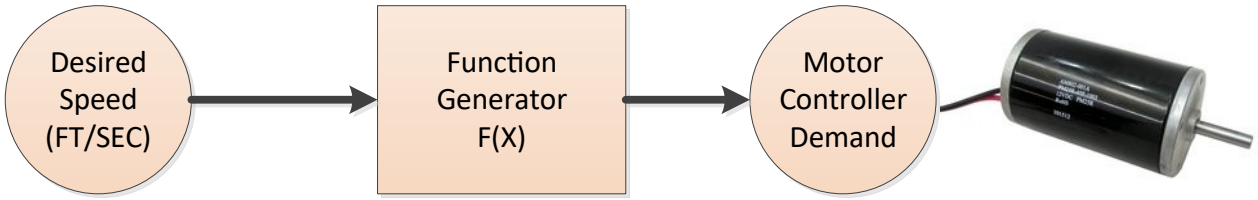
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What is “Analog Control”

- **Setting a specific output of an actuator, such as a motor, to achieve a desired result.**
  - Examples:
    - Pressing a car’s accelerator pedal to get the motor to drive the car a specific speed.
    - Sending a specific voltage to a motor to get the robot to drive a specific speed.
    - Sending a specific voltage to a motor to get an arm to be at a specific angle.

Open Loop Control

- **Setting the output blindly without knowing the current value of the controlled variable.**
  - Example - Driver cannot see the speed of the car when pressing the accelerator pedal.
- **Characteristics:**
  - Relies on anticipated response. Can't correct for changes (hill, low battery charge).
  - Responds fast, but not accurately.

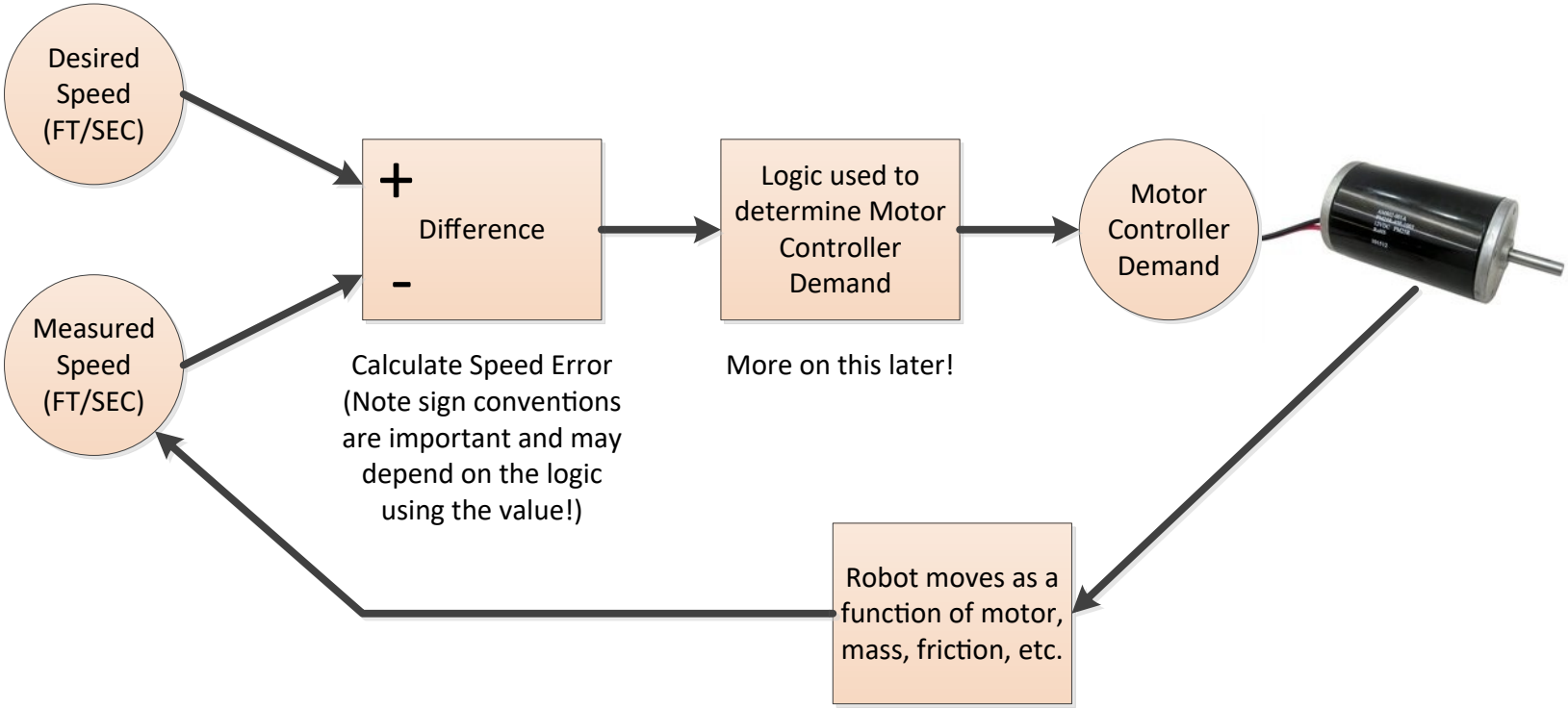


Estimates how much motor demand is required to achieve the requested speed.

Closed Loop Control

- **Output is set using feedback from sensors**
  - Example - Driver presses the accelerator pedal based on watching the speedometer.
- **Characteristics:**
  - Relies on sensed response. Has to wait for a response and then correct for error.
  - Responds accurately, but not quickly. Waiting for the system to react slows the response time. Error can be reduced to zero.

Closed Loop Control

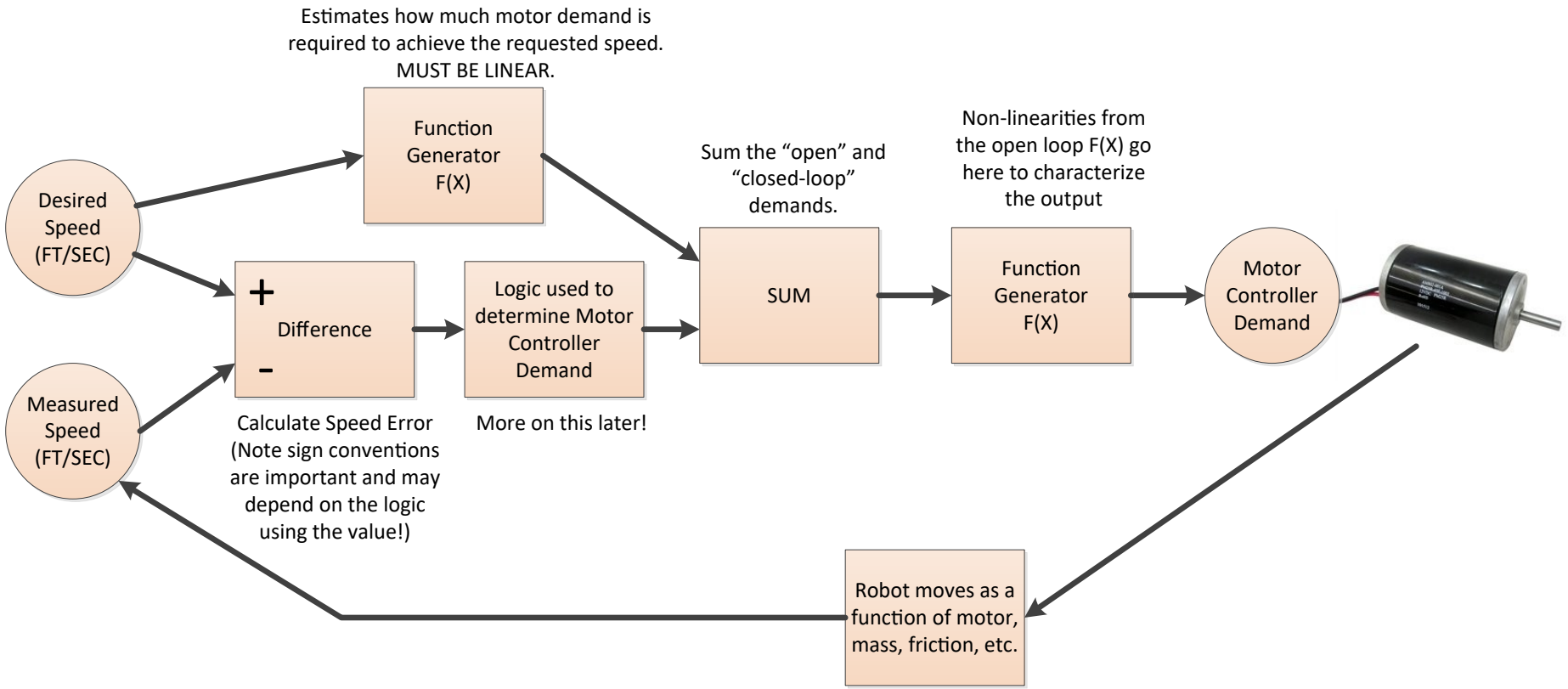


- **Motor Demand = F(Desired Speed – Actual Speed)**
  - More on the “Function” later

Closed Loop Control with Feedforward

- **Combines both Open and Closed loop concepts to achieve a fast and accurate response.**
  - Example – Driver pushes accelerator to a specific spot based on experience, and uses the speedometer to adjust to achieve an exact desired speed.

Closed Loop Control with Feedforward



- **Motor Demand = F(F(Desired Speed) + F(Speed Error))**

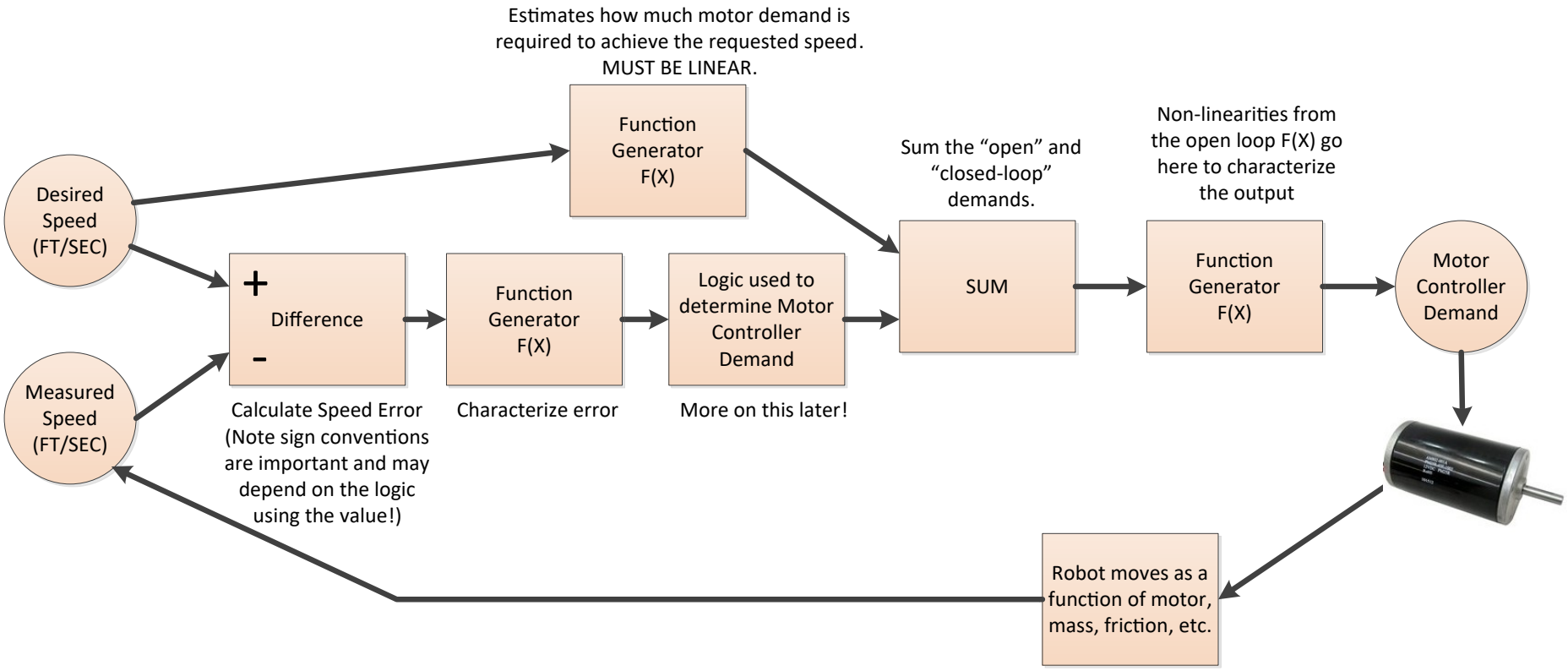
Analog Control Terminology

- **Setpoint (SP) -** Desired outcome. Example: Desired speed FT/SEC, Desired position DEG.
  - Note: Sometimes this is referred to as “demand”.
- **Process Variable (PV) -** Measured current value. Example: Actual speed FT/SEC, Actual position DEG.
- **Process Error -** Difference between setpoint and process variable. ERR = SP – PV
  - Note that the SP and PV units **must** be the same.
- **Control Output (CO) -** Value of the output to the actuator. Example: voltage demand to motor.
  - Note that the units of the CO are different and unrelated to the SP, and PV.
- **Target –** Sometimes this refers to the eventual SP. We may want to ramp the Target value to the SP over time.

Extra Credit – Items Glossed Over

- **Noisy Sensors –** Sometimes sensors have noise, which will require filtering.
- **Closed Loop Algorithms –** There are various math functions used to calculate the CO based on the process error. More on this later.
- **PV / CO Relationship Must Be Linear** – For the control algorithms to work correctly, and quickly, the same change in CO must result in the same change to PV. Otherwise the CO must be characterized to make it linear. (Note: FRC motors are linear.)
- **Derivation of Feedforward** – The straight line function of the feedforward must be derived by testing.
- **Close Loop Tuning** – Closed loop control algorithms must be tested and tuned.

Extra Credit – Full control diagram



- **Added error characterization F(x)**

Questions

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